SEONHO YOO

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RESEARCH INTERESTS

Industrial AI researcher at Korea University working on reliable machine learning, anomaly detection, and optimization-informed AI for real-world industrial systems — developing trustworthy and efficient methods for structured data and operational decision-making.

Focus areas: Anomaly Detection & Robust Predictive Modeling; Optimization-aware Learning & Decision-making; AI for Operations & Industrial Systems; Data-centric Methods for Real-world Applications.

EDUCATION

Korea University, Integrated M.S./Ph.D. in Industrial & Management Engineering (Expected) — Mar. 2027 – Present
Advisor: Prof. Seokhyun Chung. Entered the graduate track through the Integrated B.S./M.S./Ph.D. Program (Jul. 2026).

Korea University, B.S. in Industrial & Management Engineering — Mar. 2023 – Feb. 2027
Completed all credit requirements for graduation (Feb. 2026).

CAREER

SIA Lab — Apr. 2026 – Present
Research & Project Associate, Korea University — *Seoul, Korea*

- Research areas: Industrial AI & PHM, data-centric methods for real-world AI, robust predictive modeling.
- Explored foundation-model- and meta-learning-based approaches for RUL prediction across heterogeneous equipment domains.

SKI-ML Lab — Structure & Knowledge Injection into Machine Learning — Jul. 2025 – Jun. 2026
Research Intern, Seoul National University — *Seoul, Korea*

- Research areas: graph-based reasoning, multi-hop question answering, protein structure prediction.
- Designed graph-construction methods (entity linking, relation extraction) to improve multi-step inference accuracy in QA systems.
- Developed an energy-based model capturing dependencies among structured outputs, improving prediction in tasks such as protein-decoy evaluation and molecular property inference.

SAVANNA Lab — Supply Chain & Value Network Analytics — Sep. 2024 – Aug. 2025
Research Intern, Korea University — *Seoul, Korea*

- Research areas: semi-supervised learning, data augmentation, demand forecasting, optimization.
- Integrated optimization of future logistics and freight brokerage in multi-supply-chain systems — intelligent last-mile and middle-mile algorithms based on 5PL platforms.
- Location optimization of battery-swapping stations for electric two-wheelers in sustainable urban delivery (Industry–Academia Joint Technology Development, LINC 3.0).

PUBLICATIONS

Journal

- **Yoo, S.**, Joo, S., Ko, K.*, Jeong, T.* (2026). TriMa: Tri-strategy Margin-filtered Pseudo-labeling for Battery Swapping Demand Forecasting. *Journal of the Korean Institute of Industrial Engineers (KIIE).* (Accepted)

Patent

- Method and System for Optimal Location Selection of Battery Swapping Stations for Electric Two-Wheelers. Korean Intellectual Property Office (KIPO), No. 10-2025-0038793 (filed).

* Corresponding author.

PRESENTATIONS

- Ko, K., Cha, H., Kim, T., **Yoo, S.**, Jeong, T.* (2025). A Study on Multilateral Transportation Matching and Equilibrium Price Determination in a 5PL Logistics Platform Environment. Fall Conference, KIIE, Daejeon, Korea.
- **Yoo, S.**, Ko, K., Kim, T., Jeong, T.* (2025). Improving Electric Two-Wheeler Battery Swap Demand Forecasting via Clustering and Data Augmentation. Spring Conference, Korean Society of Supply Chain Management (SCM), Seoul, Korea.
- Choi, M., Ko, K., Kim, T., **Yoo, S.**, Joo, S., Jeong, T.* (2024). Location Optimization of Battery Swapping Stations for Electric Two-Wheelers in Sustainable Urban Delivery Services. Fall Conference, Korean Society of SCM, Seoul, Korea.
- Kim, T., Ko, K., Choi, M., **Yoo, S.**, Joo, S., Jeong, T.* (2024). Data-Driven Location Optimization of Battery Swapping Stations for Electric Two-Wheelers in Urban Delivery Services. Fall Conference, KIIE, Seoul, Korea.

COMPETITIONS

- **Multilateral Transportation Matching in a 5PL Logistics Platform** considering robust equilibrium price determination — Fall Conference, Korean Society of Logistics (KSCL), 2025. *Encouragement Prize.* [Yoo, S., Kim, T.]
- **Paperclip — A Tone-Sensitive Writing Assistant** for clear and culturally appropriate messaging — Edge AI Developer Hackathon (Qualcomm), 2025. *Finalist.* [repository]
- **RL-Based Volume-Efficiency Analysis in 3D Bin Packing** with stack and conveyor variables — Fall Conference, KIIE, 2024. *Bronze Prize.*
- **School-Age Population-Based Optimization of Education Facility Locations** (user-weighted LP with CFLP) — FIELD CAMP Competition, KIIE, 2023. *Excellence Prize.*

AWARDS

- Finalist, Edge AI Developer Hackathon "Innovating On-Device Intelligence," Qualcomm 2025
- Encouragement Prize, Project Competition, Korean Society of Logistics (KSCL) 2025
- Bronze Prize, 20th University Students Industrial Engineering Project Competition, KIIE 2024
- Excellence Prize, KU–Yonsei Joint Industrial Engineering Academic Exchange Festival 2024
- Excellence Prize, FIELD CAMP Competition (Public Education Facility Location Optimization), KIIE 2023
- Excellence Prize, 67th Korea National Science Exhibition, Ministry of Trade, Industry and Energy 2021

HONORS

- Presidential Science Scholarship, Ministry of Science and ICT, Korea 2025
- Sang-Ah Scholarship Foundation, Scholarship Recipient 2024
- Academic Excellence Award, Korea University 2023, 2024

UNDERGRADUATE

Activities

WeTIE — Institute for Industrial Engineering, Korea University Mar. 2023 – Present
Member & Executive Member, Education Department

- Active member (11th–15th cohorts); executive member of the Education Division (15th cohort).

- Led academic paper-review sessions (14th cohort) on social entrepreneurship and impact evaluation; organized hands-on machine-learning education sessions (15th cohort).

SK LOOKIE, Korea University — Social Venture Founders' Association Sep. 2023 – Feb. 2025
Member & President (2024)

- Team Leader, Startup Project (Fall 2023); led overall club operations in 2024 (strategic planning, member engagement, external partnerships).

Student Success Center, Korea University May 2024 – Dec. 2024
Data Analytics Team Member

- Analyzed student-participation data to derive insights on program effectiveness and engagement trends.

B.D.A — Institute for Big Data Analysis Mar. 2024 – Aug. 2024
Member, Data Analytics & Modeling Track

- Selected as an Outstanding Member (Top 1%) of the 8th cohort.

Projects

- **Regional Festival Marketing Strategy via Scientific Analysis** — developed the Festival Success Index (FSI); Industry–Academia collaboration with KT Big Data Center.
- **AI Algorithm Competition** — anomaly detection in water-supply networks (machine learning).
- **College Basketball Analytics** — winning-factor analysis and outcome prediction (machine learning).
- **Automated IPC Technology Classification** — patent classification via AI and NLP (deep learning).
- **University Timetable Optimization** — optimal timetable design for incoming students (optimization).

Others

- Vice President, 41st Student Council, Department of Industrial & Management Engineering, Korea University.
- Completed: Intellectual Property Camp (Dept. of IME, KU, 2024); Creative Challenger Program (KU, 2023); Codeit University Coding Bootcamp, 12th Cohort (2023).
- Certification: Advanced Data Analytics Semi-Professional (ADsP).